WELCOME TO

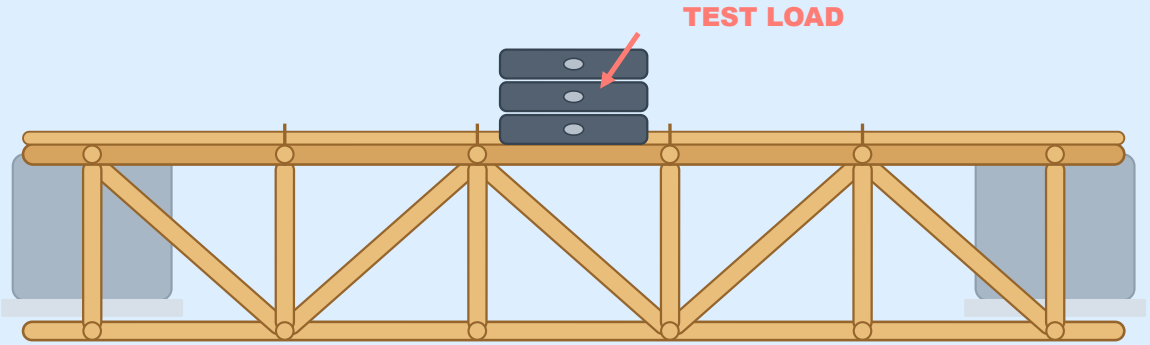
TECHIDS!

TODAY'S CHALLENGE

Build a Bridge!

Design a structure that can span a gap and carry a load.

Learn. Build. Solve.



MIDDLE SCHOOL EDITION

10–12 INCH GAP

Engineering solves problems.

Bridges connect places that would otherwise be hard or unsafe to cross.

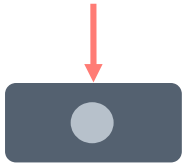
01



CONNECT SIDES

Cross a river, road, valley, or other gap.

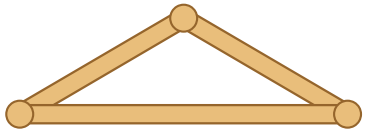
02



CARRY A LOAD

Support people, vehicles, and materials.

03



STAY STABLE

Hold the load without bending too far.

DESIGN QUESTION

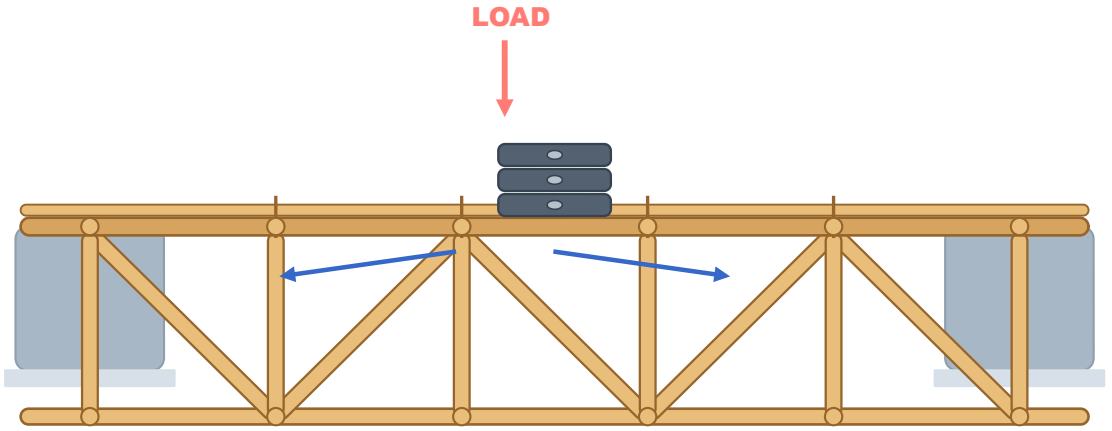
How can a small number of wooden sticks carry a load across a 10–12 inch gap?

Think like an engineer: solve the problem, then prove the design works.

TECHIDS

How does a bridge carry a load?

A load pushes down. The bridge structure transfers forces toward the supports.



FORCE PATH TO SUPPORTS

COMPRESSION

A member is squeezed or pushed together.

TENSION

A member is pulled or stretched.

BUILD TAKEAWAY: stiff deck • strong joints • forces guided to both supports

Why do engineers use triangles?

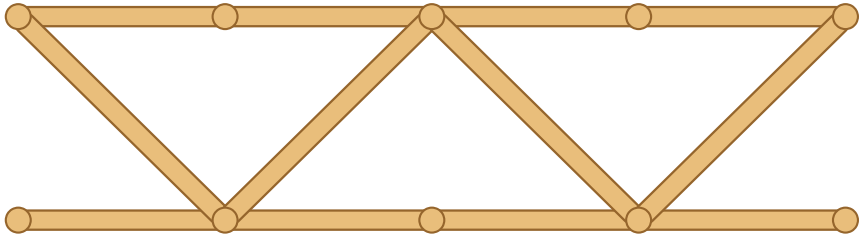
Triangles help a bridge keep its shape when forces push and pull on it.

WITHOUT A DIAGONAL



A rectangle can shift into a slanted shape.

WITH TRIANGULAR TRUSSES



TRIANGLES RESIST RACKING

ENGINEERING TIP: Build two matching side trusses with repeated triangles. Glue the joints, then connect the sides with crosspieces and a deck.

YOUR MISSION

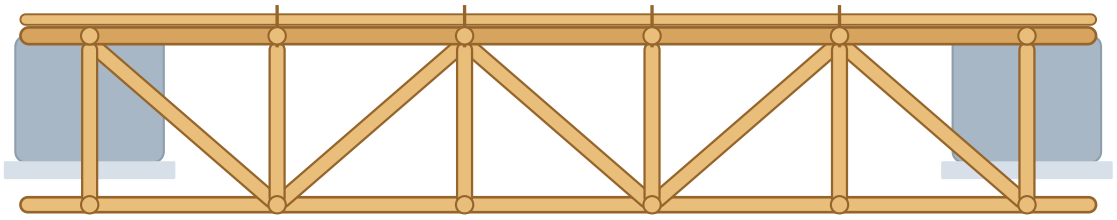
TECHIDS

Build a craft-stick bridge.

Your team has a limited materials budget. Design first, then build.

THE GOAL

Span the gap. Carry the load.



10–12 INCH CLEAR GAP

SUPPORT

SUPPORT

THE RULES

01 — 25 MINUTES
Plan, build, and stop when time is up.

02 — FAIR MATERIALS
Same craft-stick count and glue allowance per team.

03 — WHITE GLUE ONLY
Use Elmer's white glue at the joints.

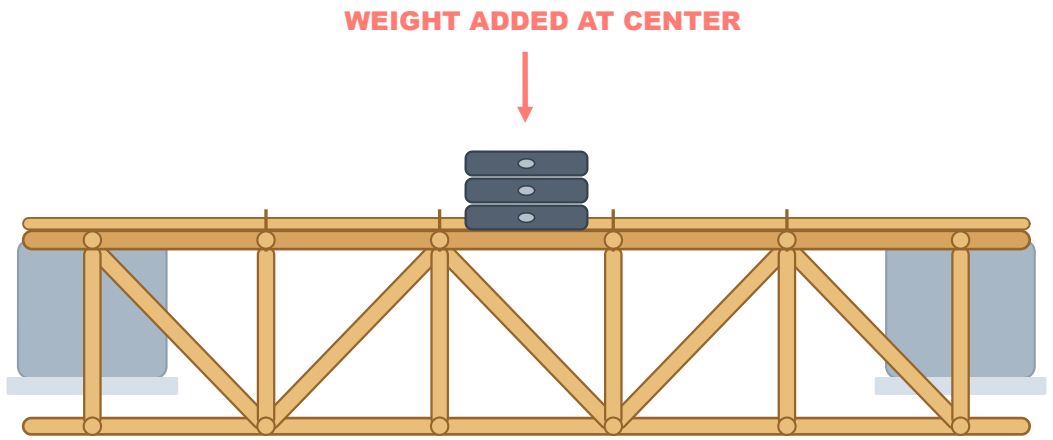
04 — SPAN BOTH SIDES
Rest on the two raised supports; nothing underneath.

BUILD SMART

TRIANGLES • STRONG GLUE JOINTS • CONNECT BOTH SIDES • LOAD PATH TO SUPPORTS

Official load test: how strong is each bridge?

Techids instructors add the weights and track the results. Students observe the design.



SAME TEST FOR EVERY TEAM

1. Same 10–12 inch gap
2. Same weight increments
3. Weight placed at the center
4. Stop at the agreed failure point
5. Instructor records maximum load and what failed

INSTRUCTORS HANDLE WEIGHTS

OBSERVE LIKE AN ENGINEER

Where did it bend? Which joint moved first? What would you change?

What did we learn about strong bridges?

A strong design uses shape, connections, and a clear path for the load.

01

TRIANGLES & TRUSSES

Repeated triangles help a frame resist changing shape and make the structure stiffer.

02

COMPRESSION & TENSION

Bridge members can be squeezed or pulled. A good design guides those forces through the frame.

03

LOAD PATH & DISTRIBUTION

The deck transfers weight into the truss, then toward both supports and the ground.

04

JOINTS & ITERATION

Weak glue joints can undo a smart shape. Test results show what to improve next.

THE ENGINEERING CYCLE

DEFINE → DESIGN → BUILD → TEST → IMPROVE

A stronger bridge is not just made with more sticks — it is designed so the sticks work together.